TIGHT HARDY-TO-BERGMAN INCLUSIONS ON THE UNIT BALL

ØISTEIN SØVIK

Evidence is labelled Sketch: this proof has not been posted on arXiv or peer reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> Conjecture 22(b). Then $H^p \subset A^q_\alpha$ is a tight fitting if and only if
>
> $$p < q \quad \text{and} \quad nq = (n+1+\alpha)p.$$
>
> Furthermore, $\|f\|_{A^q_\alpha} = \|f\|_{H^p}$ if and only if $f(z) = c/(1-\langle z,a\rangle)^{2n/p}$, where $c \in \mathbb{C}$ and $a \in \mathbb{B}^n$.

Here $0 < p, q < \infty$, $\alpha > -1$, and $n \geq 1$. Set $M = n + 1 + \alpha > n$. A tight fitting is a proper, contractive, noncompact inclusion. Source: Bao, Ma, Yan, and Zhu, "Embedding and compact embedding between Bergman and Hardy spaces," arXiv:2502.08406v1, Conjecture 22(b), p. 19.

Answered: yes. Evidence: Sketch.

Conditional on: Li–Su, *Contraction property on complex hyperbolic ball*, arXiv:2411.01911v2, Assumption 1.4 and Theorem 4.1; the geometric hypothesis is claimed to follow from Kalaj, *The Gromov–Ros conjecture for rank-one symmetric spaces of noncompact type*, arXiv:2607.20055v3.

The geometric hypothesis is that geodesic balls minimize perimeter at fixed volume in complex hyperbolic space. Conditional on that hypothesis, tight fitting holds exactly when $p < q$ and $nq = (n+1+\alpha)p$, and equality occurs exactly for the stated kernel family $c(1-\langle z,a\rangle)^{-2n/p}$.

Proof. First take $nq = Mp$ and put $r = p/n = q/M > 0$. Then $M > n$ gives $q > p$. For $f \neq 0$, normalize $\|f\|_{\mathcal{H}^p} = 1$, and define

$$u(z) = |f(z)|^r(1-|z|^2), \qquad \mathrm{d}v_g = \frac{\mathrm{d}v}{(1-|z|^2)^{n+1}}, \qquad \mu_f(t) = v_g(\{u > t\}).$$

Li and Su's Theorem 4.1, under Assumption 1.4, give

$$\mu_f(t) \leq \mu_1(t), \qquad \mu_1(t) = \begin{cases} (t^{-1}-1)^n, & 0 < t < 1, \\ 0, & t \geq 1. \end{cases} \tag{1}$$

Here the model is $f = 1$. The normalization is literal: Li–Su's invariant Bergman parameter is M, so their density $c_M(1-|z|^2)^M\,\mathrm{d}v_g$ equals the density defining A^q_α above. By layer cake and (1),

$$\|f\|^q_{A^q_\alpha} = c_M M \int_0^\infty t^{M-1}\mu_f(t)\,\mathrm{d}t \leq c_M M \int_0^1 t^{M-1}(t^{-1}-1)^n\,\mathrm{d}t = 1. \tag{2}$$

The last integral is the beta integral $B(M-n, n+1)$, and is finite since $M > n$. This proves contraction with norm one, since constants attain it.

Suppose equality holds in (2). The nonnegative difference in (1) then has zero integral against the strictly positive weight $t^{M-1}\,\mathrm{d}t$ on $(0,1)$; hence $\mu_f = \mu_1$ almost everywhere there. The sharp point-evaluation estimate (1.2) of Li and Su gives $u \leq 1$. It also gives uniform boundary decay for a fixed f, as follows. Approximate f in H^p by a polynomial P; density is stated immediately after their estimate (1.2). Point evaluation applied to $f - P$ yields

$$(1-|z|^2)|f(z)-P(z)|^{p/n} \leq \|f-P\|^{p/n}_{H^p}.$$

Date: 13 October 2026.

Together with $|x+y|^{p/n} \le C_{p/n}(|x|^{p/n}+|y|^{p/n})$ and the boundedness of P, this implies $u(z)\to 0$ uniformly as $|z|\to 1$. Consequently u attains its maximum in $\mathbb{B}_n$. If its maximum were $m<1$, then $\mu_f(t)=0<\mu_1(t)$ throughout $(m,1)$, contradicting the almost-everywhere equality. There is therefore $a\in\mathbb{B}_n$ such that

$$|f(a)|^p(1-|a|^2)^n = 1. \tag{3}$$

Let φ_a be the standard involutive ball automorphism interchanging 0 and a, and set

$$F(w) = \frac{(1-|a|^2)^{n/p}}{(1-\langle w,a\rangle)^{2n/p}} f(\varphi_a(w)).$$

All powers use the holomorphic logarithm on the right half-plane. This transform is an H^p isometry. Indeed its boundary multiplier to the power p is the sphere Jacobian

$$J_{\varphi_a}(\zeta) = \frac{(1-|a|^2)^n}{|1-\langle\zeta,a\rangle|^{2n}},$$

so boundary change of variables preserves the L^p integral. This also holds for arbitrary H^p functions by polynomial density: apply the identity first to polynomials, then use point evaluation and completeness of H^p to pass to the limit. Thus $\|F\|_{H^p}=|F(0)|=1$ by (3).

For every $0<\rho<1$, subharmonicity and the Hardy norm now imply

$$1=|F(0)|^p \le \int_{\mathbb{S}_n} |F(\rho\zeta)|^p \,\mathrm{d}\sigma(\zeta) \le \|F\|_{H^p}^p = 1.$$

The equality of every centered mean implies that the continuous subharmonic function $|F|^p$ is harmonic: its positive Riesz measure has zero integral against every ball Green kernel. Near 0 the function F is nonzero, and there

$$\Delta|F|^p = p^2|F|^{p-2}\sum_{j=1}^n \Big|\frac{\partial F}{\partial w_j}\Big|^2.$$

Hence all derivatives vanish near 0, so F is constant by holomorphic continuation. Using $\varphi_a^2=\mathrm{id}$ and

$$1-\langle\varphi_a(z),a\rangle = \frac{1-|a|^2}{1-\langle z,a\rangle}$$

recovers exactly the proposed kernel family for f. Conversely these kernels are transforms of constants. The identity

$$1-|\varphi_a(z)|^2 = \frac{(1-|a|^2)(1-|z|^2)}{|1-\langle z,a\rangle|^2}$$

and the invariance of $\mathrm{d}v_g$ show that the same transform is an A^q_α isometry when $Mq^{-1}=np^{-1}$. Thus every proposed kernel attains equality in the two norms; $f=0$ is included by $c=0$.

Finally, Theorem A(b), (c) of Bao, Ma, Yan, and Zhu show that for $p<q$ the inclusion exists precisely when $n/p\le M/q$, and it is noncompact precisely at equality. Their Theorem B(b), (c) makes all the inclusions with $p\ge q$ and $\alpha>-1$ compact. On the critical line the reverse inclusion $A^q_\alpha\subset H^p$ is impossible by their Theorem B(a), since it would require $\alpha\le -1$ (or $\alpha<-1$ when $p>2$). Hence the critical inclusion is proper. These statements prove both directions of the asserted tight-fitting criterion. □